GN3 Domain Alignment in SALT

Latent-Space Domain Alignment with Sinkhorn Optimal Transport

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Why Domain Alignment?

Goal: reduce MC–data scale-factor uncertainty during calibration.

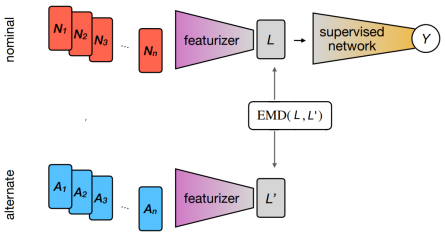
- We expect some raw MC tagging performance to drop: the tagger should not profit from MC mis-modeling.
- The target is a representation that extracts maximal flavor information while being less sensitive to MC/data domain effects.
- Desired tradeoff: lower raw MC performance but smaller SF uncertainty; both effects compound but could help downstream analyses in the long run (to be determined).

Train for flavor discrimination, not simulation artifacts.

Minimal Domain-Alignment Baseline

MACK/ROCK-style implementation in SALT

- shared GN3/SALT transformer backbone for nominal and alternate samples
- nominal flavor labels train the classification head
- `source_label` domain labels split samples for Sinkhorn optimal transport
- each jet is represented by `global_rep` for both classification and alignment
- Optimal transport, approximated by Sinkhorn loss, aligns nominal and alternate latent distributions with batch-level EMD



Minimal domain-alignment model.

What is actually optimized?

$$L_{\text{total}} = L_{\text{CE,nominal}} + \lambda D_{\text{Sinkhorn}}(z_{\text{nominal}}, z_{\text{alternate}})$$

Contributes to classification loss

- nominal samples only
- standard jet-flavor cross-entropy
- alternate labels are ignored for CE

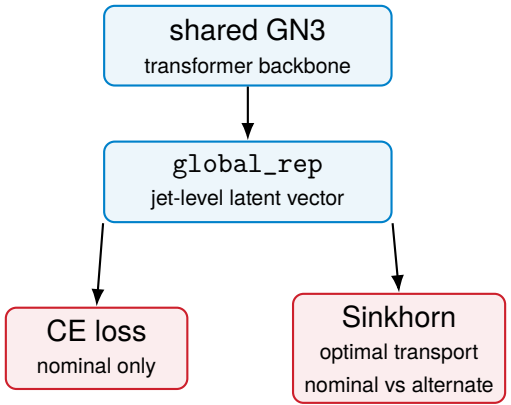
Contributes to alignment loss

- nominal embeddings
- alternate embeddings
- Optimal transport, approximated by Sinkhorn loss, compares the two latent distributions

The alignment term acts on the representation, not on the classifier logits.

Implementation: three SALT hooks

- **Config:** the YAML config instantiates the Sinkhorn alignment loss, sets its weight λ, and uses `source_label` to distinguish nominal from alternate jets.
- **CE masking:** in `modelwrapper.py`, every jet runs through the same GN3 backbone; the wrapper masks the supervised flavor CE so only nominal jets train the classifier head.
- **Alignment loss:** `SinkhornDomainAlignmentLoss.py` takes `preds["global_rep"]`, splits embeddings by `source_label`, and adds an EMD-like distance between the two latent batches.



Discussion points

- **Choosing λ:** pseudo-data gives a verifiable optimum; real data does not.
- **Batch-level Sinkhorn:** are minibatch domain distances stable enough to replace jet-level EMD?
- **Sample balance:** do nominal and alternate batches need similar jet flavor mixtures for the alignment to be meaningful?
- **Calibration endpoint:** train custom GN3, rerun the full chain, and measure changes in ϵ_{data} and SF uncertainty.

Main question: does optimal transport reduce SF uncertainty, and is it worth lower raw MC performance?

Get started: gitlab.cern.ch/lvaughan/domain-alignment-toolkit